

ROHIT SHINDE

GenAI Engineer | LLMs & RAG | Agentic AI | Python Full Stack | MLOps

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PROFESSIONAL SUMMARY

GenAI Engineer and Python Full-Stack Developer with 4+ years of experience building AI-powered products across the complete software lifecycle—from requirements and architecture to development, deployment, optimization, and monitoring. Specialized in LLMs, RAG, Agentic AI, NLP, LLM fine-tuning, multimodal GenAI, and MLOps/LLMOps, with hands-on experience using PyTorch, Hugging Face, LangChain, LlamaIndex, FastAPI, Django, Docker, and GCP. Experienced in fine-tuning open-source models using LoRA, QLoRA, and progressive layer unfreezing, designing production-grade RAG systems, optimizing LLM inference, and deploying AI microservices to cloud environments.

Core strengths: AI product development · LLM engineering · RAG architecture · model fine-tuning · AI agents · MLOps/LLMOps · Python backend engineering · cloud deployment

TECHNICAL SKILLS

Programming & Data	Python 3.x, Pandas, NumPy, SciPy, Scikit-learn, NLTK, spaCy, Matplotlib, Plotly
Deep Learning & NLP	PyTorch, TensorFlow/Keras, Hugging Face Transformers, ONNX, TensorRT, NLP, Computer Vision
Generative AI & LLMs	GPT-4, Gemma 3B, Llama, Mistral, Prompt Engineering, Function Calling, Agentic AI, Multimodal AI
RAG & AI Frameworks	LangChain, LlamaIndex, FAISS, ChromaDB, Milvus, Document Processing, Embeddings, Retrieval Pipelines
Fine-Tuning & LLMOps	LoRA, QLoRA, PEFT, vLLM, MLflow, Weights & Biases, Experiment Tracking, Model Versioning, Model Monitoring
Backend & APIs	FastAPI, Django, Flask, REST APIs, Async Python, JWT, 2FA, Microservices
Cloud & DevOps	GCP (Vertex AI, Cloud Run, Cloud Functions), AWS, Docker, CI/CD, Git
Generative Vision	Stable Diffusion, SDXL, SD3, ControlNet, LoRA, ComfyUI, VAE, UNet

PROFESSIONAL EXPERIENCE

AI / GenAI Engineer — LLM Fine-Tuning, RAG & Agentic AI

EduAI Hub — Pune, India | Dec 2025 – Jun 2026

- Led development of the EduAI SLM, fine-tuning Gemma 3B using LoRA, QLoRA, and progressive layer unfreezing, improving domain-specific Q&A accuracy by ~38%.
- Architected production RAG pipelines using LlamaIndex and ChromaDB, grounding responses in curriculum PDFs and reducing hallucinations by ~42%.
- Reduced LLM inference latency by ~40% through QLoRA, PEFT, and vLLM-based serving optimization.
- Established an LLMOps workflow using MLflow and Weights & Biases for experiment tracking, versioning, and evaluation.
- Developed and deployed GenAI microservices using FastAPI, Docker, and GCP Cloud Run, supported by CI/CD pipelines.
- Designed multimodal content-generation workflows integrating Stable Diffusion SDXL and ControlNet for educational visuals; mentored junior engineers in RAG, fine-tuning, and MLOps.

GenAI & Machine Learning Developer — Freelance

Independent Projects — Remote | Jan 2024 – Present

- Delivered end-to-end GenAI and ML solutions for international clients, including LLM chatbots, RAG applications, AI analytics systems, and agentic workflows.
- Designed AI solutions from requirements and architecture through implementation, testing, deployment, and optimization.
- Automated batch AI-generation workflows using Python and FastAPI, reducing manual operational effort by ~70%.
- Developed scalable backend APIs and AI services, integrating LLM and diffusion-model pipelines for multimodal applications.

Founder / Python Full-Stack & AI Engineer

Stones Web Services — Pune, India | Jun 2021 – Present

- Founded and delivered technology solutions for 15+ startups, covering AI engineering, backend development, APIs, and full-stack product development.
- Designed and developed production applications using Django, FastAPI, React, Python, and REST APIs, incorporating OpenAI APIs, LangChain, RAG, and diffusion models.
- Collaborated directly with product teams and founders to translate business requirements into practical AI/ML solutions.
- Deployed production applications and AI services on GCP using Docker and CI/CD; led technical decisions across architecture, development, and iteration.

R&D Specialist — Artificial Intelligence

Dr. D. Y. Patil School of Science & Technology — Pune, India | Nov 2024 – Apr 2025

- Researched and implemented Computer Vision, NLP, and Machine Learning techniques for an AI-powered Electronic Voting System, including real-time fraud detection.
- Researched diffusion-based synthetic data generation for security-model training and contributed to system architecture involving facial recognition and analytics.

Technical Trainer — Python & Full Stack

Teknowell EduTech — Pune, India | Aug 2024 – Nov 2024

- Trained 50+ students in Python, Django, Flask, React, REST APIs, Git, deployment, and software development practices.
- Mentored students on project development, debugging, deployment, and technical interview preparation.

SELECTED AI PROJECTS

EduAI SLM — Gemma 3B Fine-Tuning + RAG

PyTorch · Hugging Face · LlamaIndex · ChromaDB · FastAPI · GCP · MLflow

- Fine-tuned Gemma 3B on educational datasets using LoRA, QLoRA, and progressive layer unfreezing, improving domain-specific Q&A accuracy by ~38%.
- Built a RAG architecture using LlamaIndex + ChromaDB to ground responses in curriculum documents, reducing hallucinations by ~42%.
- Implemented streaming inference and production serving on GCP Cloud Run, with an MLflow-based LLMOps workflow for tracking and evaluation.

Multi-Modal Educational Content Pipeline

Stable Diffusion SDXL · ControlNet · LoRA · ComfyUI · FastAPI

- Developed a multimodal GenAI pipeline generating educational diagrams and visual content from lesson prompts.
- Integrated SDXL, ControlNet, and LoRA for controlled image generation; exposed workflows through FastAPI microservices.
- Designed the pipeline for integration with an existing Learning Management System.

AI-Powered Electronic Voting System

Python · Django · FastAPI · Computer Vision · Machine Learning

- Developed a secure electronic voting platform incorporating facial recognition, 2FA, real-time analytics, and ML-based fraud detection.
- Designed backend services using Django and FastAPI; applied computer vision and ML techniques for identity verification and anomaly detection.

EDUCATION

B.Tech — Artificial Intelligence & Data Science

Dr. D. Y. Patil Vidyapeeth, Pune | 2021 – 2025 | CGPA: 9.45 / 10

CERTIFICATIONS

- Google Advanced Data Analytics Professional Certificate — Google
- The Power of Statistics — Coursera

ADDITIONAL INFORMATION

Experience	4+ years — Python Full Stack & AI/ML GenAI/LLM/RAG: 1.5+ yrs Fine-Tuning/LoRA/Diffusion: 1.5+ yrs
Cloud	GCP — Vertex AI, Cloud Run, Cloud Functions AWS — Working Knowledge
Availability	Immediate Location: Pune, India Work Preference: On-site / Hybrid / Remote